



# Process intensification of a catalytic-wall Taylor-Couette reactor through unconventional modulation of its angular speed

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## ABSTRACT

This study proposes a non-conventional operation of a Taylor-Couette reactor by introducing a periodic flow perturbation as a modulated angular velocity of its inner cylinder. The applicability of this type of waveform has yet to be explored in Taylor-Couette reactors as a means to enhance mass transfer phenomena in reactive systems. A multiphysics numerical study was carried out considering a mass-transfer limited system with a catalytic reaction at the outer cylinder boundary of the reactor while applying different modulations of the inner cylinder angular speed. Results showed that a modular signal could yield conversions similar to a constant angular-speed operation. However, the use of modulating signals brings two essential benefits. First, it enhances mass transfer, which yields higher conversions by dynamically changing the flow patterns in the reactor. This improvement is demonstrated and discussed in terms of a dynamic mixing index, which accounts for the formation and abrupt disruption of Taylor vortices in the reactor. Second and more importantly, this type of operation leads to an overall reduction in the average electrical power required to drive the system (~25 % reduction). The present study opens the possibility of using intelligent control strategies to optimize reactions and intensify conventional systems with non-conventional operation modes.

## Nomenclature

|               |                                    |
|---------------|------------------------------------|
| $c$           | Concentration                      |
| $c_0$         | Inlet concentration                |
| $c_{lim}$     | Limiting reactant concentration    |
| $d$           | Annular gap width                  |
| $d_i$         | Inner cylinder diameter            |
| $d_o$         | Outer cylinder diameter            |
| $e_{mot}$     | Motor armature voltage             |
| $i_{mot}$     | Motor armature current             |
| $p$           | Pressure                           |
| $r$           | Radial axis                        |
| $r_i$         | Inner cylinder radius              |
| $u_{ax}$      | Axial fluid velocity component     |
| $v$           | Azimuthal fluid velocity component |
| $C$           | Normalized concentration           |
| $C_{cup}$     | Mixing cup concentration           |
| $D$           | Angular speed waveform duty cycle  |
| $\mathcal{D}$ | Diffusion coefficient              |
| $H$           | Height of the reactor              |

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## Nomenclature (continued)

|            |                                             |
|------------|---------------------------------------------|
| $I_{mix}$  | Mixing index                                |
| $J$        | Diffusive flux                              |
| $J_{tot}$  | Moment of inertia of rotating parts         |
| $K_e$      | Motor back electromotive force constant     |
| $K_t$      | Motor torque constant                       |
| $L$        | Motor winding inductance                    |
| $P_e$      | Electrical power                            |
| $P_{mot}$  | Motor power                                 |
| $R$        | Motor winding resistance                    |
| $R_j$      | Total rate of reaction                      |
| $Q_0$      | Constant flow rate                          |
| $T$        | Angular speed waveform period               |
| $T_+$      | Angular speed waveform positive semi-period |
| $T_{bear}$ | Bearing friction torque                     |
| $T_{drag}$ | Drag torque                                 |
| $T_{in}$   | Inertial torque                             |
| $T_{mot}$  | Motor torque                                |
| $T_0$      | Dry friction torque                         |
| $z$        | Axial axis                                  |

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## Nomenclature (continued)

|                              |                                     |
|------------------------------|-------------------------------------|
| <b>F</b>                     | Volume force vector                 |
| <b>I</b>                     | Identity matrix                     |
| <b>J</b>                     | Mass flux vector                    |
| <b>K</b>                     | Viscous stress tensor               |
| <b>u</b>                     | Velocity vector                     |
| <b>n</b>                     | Surface normal vector               |
| $\alpha$                     | Heat transfer coefficient           |
| $\beta_1$                    | Drag torque linear coefficient      |
| $\beta_2$                    | Drag torque quadratic coefficient   |
| $\eta_{cont}$                | Motor controller efficiency         |
| $\lambda$                    | Inner-to-outer diameter ratio       |
| $\lambda_l$                  | Thermal conductivity                |
| $\rho$                       | Density                             |
| $\mu$                        | Dynamic viscosity                   |
| $\nu$                        | Kinematic viscosity                 |
| $\omega$                     | Angular speed                       |
| $\omega_0$                   | Angular speed bias                  |
| $\omega_b$                   | Angular speed average value         |
| $\Delta\omega$               | Angular speed amplitude             |
| $\Gamma$                     | Aspect ratio                        |
| <b>Dimensionless numbers</b> |                                     |
| $Pe$                         | Péclet number                       |
| $Re_{ax}$                    | Axial Reynolds number               |
| $Re_{rot}$                   | Rotational Reynolds number          |
| $Re_{rot,cr}$                | Critical rotational Reynolds number |
| $Re_{rot}^*$                 | Rotational Reynolds number ratio    |
| $Sc$                         | Schmidt number                      |
| $Ta$                         | Taylor number                       |
| <b>Acronyms</b>              |                                     |
| ADF                          | Axially displaced flow              |
| LCF                          | Laminar Couette flow                |
| MWVF                         | Modulated wavy vortex flow          |
| PI                           | Process intensification             |
| TCR                          | Taylor-Couette reactor              |
| TTVF                         | Turbulent Taylor vortex flow        |
| TVF                          | Taylor vortex flow                  |
| WVF                          | Wavy vortex flow                    |

## 1. Introduction

Advancing material and energy efficiencies of chemical processes could play a critical role in addressing current environmental challenges [1]. Process Intensification (PI) has been recognized as a powerful strategy that has guided the novel design of chemical reaction equipment and methods that significantly minimize environmental impact, reducing energy and material consumption while decreasing the overall risk of operation and capital investment of the chemical process [2]. Many industries have benefited from intensified systems by implementing PI core approaches, e.g., structure, energy, synergy, and time [3]. For example, miniaturized continuous flow systems have reduced  $10\times$  the overall material and energy expenditure for the manufacturing of fine chemicals [4], energy-harvesting materials [5], pharmaceuticals [6], high-functional materials [7], and bioprocesses [8]. However, the adoption of intensified systems can be perceived as challenging due to their complex design, operation, and scale-up strategies. Therefore, most of the research in PI focuses on the novel usage of established tools rather than enhancing their intensification capabilities [9].

Taylor-Couette Reactors (TCR) are a promising intensification device for mass-transfer limited systems due to their capacity to fine-tune shear stresses and residence times, uniform mixing, and high surface-to-volume ratios. The TCR is a cylindrical shell in which an inner cylinder is inserted, creating an annular gap. While the cylinder and the shell could rotate with the angular velocity ( $\omega$ ), the shell is fixed in most applications. Based on the cylinders' rotational speed and dimensionless relations, i.e., aspect ratio ( $\Gamma = H/d$ ), inner-to-outer diameter ratio ( $\lambda = d_i/d_o$ ), and gap width ( $d = [d_o - d_i]/2$ ), mixing conditions are established independently from the axial flow ( $u_{ax}$ ).

Several dimensionless numbers have been defined to parameterize the behavior of the TCR. The rotational Reynolds number

$$Re_{rot} = \frac{\omega r_i d}{\nu} \quad (1)$$

and the Taylor number

$$Ta = \frac{2\lambda^2 d^4}{1 - \lambda^2} \left(\frac{\omega}{\nu}\right)^2 \quad (2)$$

are typically used to characterize the transition to different flow regimens. They link centrifugal and viscous forces. Increasing both values for a given TCR design indicates higher turbulence, leading to good mixing properties and plug-flow behavior [10]. The axial Reynolds number

$$Re_{ax} = \frac{u_{ax} d}{\nu} \quad (3)$$

relates to the axial velocity; at sufficiently large values ( $>10$ ), it could distort the flow pattern helically [11]. For systems where mass transfer is critical, the Schmidt number

$$Sc = \frac{\nu}{D} \quad (4)$$

and the Péclet number

$$Pe = \frac{vd}{D} \quad (5)$$

provide insight into how mass is transferred due to viscous momentum and molecular mass transport.

In a TCR, the interplay between geometric proportions (gap and inner cylinder), fluid properties, and operational parameters like angular speed gives rise to complex fluid dynamics. Consequently, a significant number of studies has focused on different flow regimes, with varying mixing conditions from dispersing to highly segregated flow, and their effect on process-dependent applications [12]. These studies cover key application areas for the TCR, as seen in Fig. 1. Given the mixing properties, the TCR is ideal for homogeneous transition metal catalysis because of the narrow residence time distribution. Uniform mixing in the annular space generates a homogeneous suspension of heterogeneous particles suitable for photocatalysis. In aggregation and flocculation, achieving different conformations or size distributions is useful. Manageable hydrodynamic behavior and low shear forces allow for narrow particle size distributions and accurate control of the polymer particles. Fluid dynamic variables, TCR design, and geometry significantly impact size distribution, microstructure, and morphology for crystallization and precipitation of solids. The uniformly low shear forces that the generated vortices exert on the liquid phase are appreciated for catalyzed reactions with enzymes.

Several strategies to improve the performance of TCR have focused on altering features of the inner cylinder, such as its shape [13–15], roughness [16,17], and surface hydrophobicity [18], among others [19]. Fewer strategies have further dealt with novel process-control methods to intensify the inherent reactor capabilities [20]. Herein, a periodic flow perturbation is introduced as modulation of the angular speed of the inner cylinder without further modification of the geometrical characteristics of the TCR. Although perturbations to the angular speed have been reported previously, these studies have focused on characterizing transient states of turbulence of the TCR [21,22]. This non-conventional operation of the TCR will be analyzed to improve the performance of a mass-transfer limited system (i.e., catalytic wall TCR).

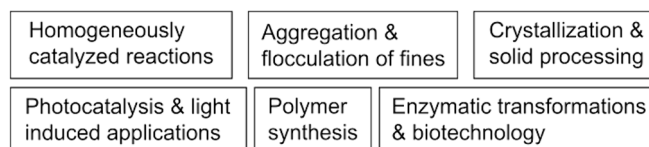


Fig. 1. Typical applications of Taylor-Couette reactors [12].

A surface response of the catalytic conversion was obtained after a grid search of the angular speed modulation parameter space. Exploration of this model led to the identification of an improved intensified system. Similar strategies where data has been a key enabler to advance the performance of intensified systems through non-conventional operation have been reported for different applications [9,23,24]. Process intensification through the data approach could potentially reduce the barrier of adopting or modifying current intensified reactive systems [25]. This opens the possibility of using intelligent control strategies enabled by data-driven analysis to optimize the conversion of reactive systems by applying suitable input perturbations. Attending this need, this work explores the periodic perturbation of the angular speed in a catalytic-wall Taylor-Couette Reactor to determine the key operational parameters that improve the reactor performance while providing insights into the fluid mechanics that enable this enhancement.

## 2. Numerical methodology

The performance of catalytic-wall TCR under the effects of an angular speed modulation was obtained by carrying out different numerical simulations using COMSOL Multiphysics. Axial symmetry was exploited to formulate a Graetz-L  v  que-type problem for laminar boundary layer mass transport. The reactor inlet is fed with a constant flow rate  $Q_0$ , whereas its inner wall is driven with an angular speed setpoint  $\omega(t)$ .

### 2.1. Geometry specifications

The reactor dimensions and the boundary conditions used in the simulation are summarized in Fig. 2 and Table 1. The aspect ratio  $\Gamma$  and the inner-to-outer diameter ratio  $\lambda$  were chosen to neglect entrance effects without sacrificing volume for the reaction and to ensure the

**Table 1**

Relevant parameters of the Taylor-Couette reactor under study.

| Parameter                  | Symbol    | Value | Unit                |
|----------------------------|-----------|-------|---------------------|
| Inlet flow rate            | $Q_0$     | 100   | mL/min              |
| Inlet concentration        | $c_0$     | 50    | mmol/m <sup>3</sup> |
| Inner cylinder diameter    | $d_i$     | 50.8  | mm                  |
| Outer cylinder diameter    | $d_o$     | 63.5  | mm                  |
| Reactor height             | $H$       | 190.5 | mm                  |
| Annulus gap                | $d$       | 6.35  | mm                  |
| Aspect ratio               | $\Gamma$  | 30    | –                   |
| Inner-to-outer diam. ratio | $\lambda$ | 0.8   | –                   |

formation of stable vortices throughout the TCR [112].

### 2.2. Problem formulation

The TCR was simulated by coupling two key modules in COMSOL Multiphysics (i.e., laminar flow and transport of diluted species). Thus, reactor fluid dynamics are governed by single-phase laminar flow equations constituted by continuity:

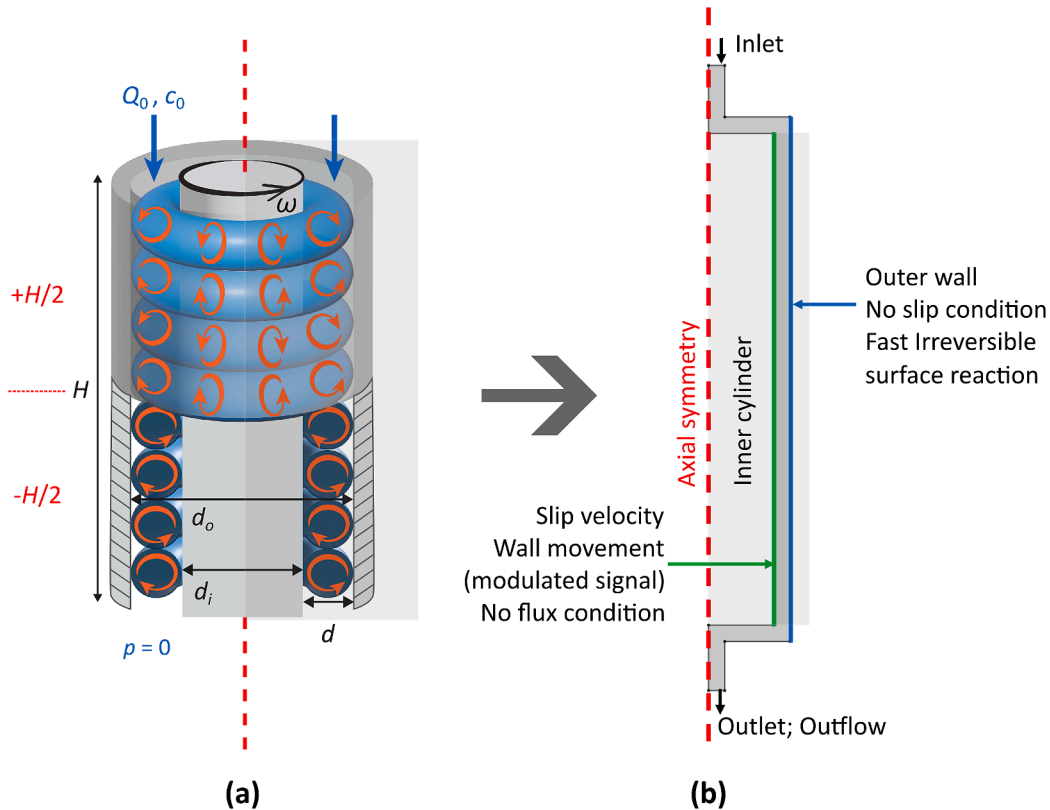
$$\rho \nabla \cdot \mathbf{u} = 0 \quad (6)$$

and momentum conservation:

$$\rho \frac{\partial \mathbf{u}}{\partial t} + \rho (\mathbf{u} \cdot \nabla) \mathbf{u} = \nabla \cdot [-p\mathbf{I} + \mathbf{K}] + \mathbf{F} \quad (7)$$

where  $\mathbf{K} = \mu (\nabla \mathbf{u} + (\nabla \mathbf{u})^T)$  is the viscous stress tensor.

These expressions present null initial conditions in all the domains:  $p = 0, \mathbf{u} = 0$ . The volumetric flow rate at the inlet of the TCR was fixed to  $Q_0$ . The fluid properties correspond to a silicon oil with  $\rho = 930 \text{ kg/m}^3$  and  $\mu = 0.01 \text{ Pa}\cdot\text{s}$ . All simulations yield  $Re_{ax} \sim 33$ . These fluid properties were chosen to represent a case study of a highly mass-transfer



**Fig. 2.** Taylor-Couette reactor. (a) Geometry characterized by inner cylinder diameter  $d_i$ , outer cylinder diameter  $d_o$ , annulus gap  $d$  and reactor height  $H$ , divided into positive and negative halves  $\pm H/2$ . The reactor is fed by inlet flow rate  $Q_0$  and concentration  $c_0$ ; its outlet pressure  $p$  is null. (b) Simulation boundary conditions represented in a model with axial symmetry.

limited reaction in liquid phase, with  $Pe \sim 3 \cdot 10^5$  and  $Sc \sim 1 \cdot 10^5$ . That is, momentum diffusivity governs mass diffusive transport to the reactive wall. Moreover, a high viscous diffusivity rate can improve the convergence of a weakly nonlinear fluid flow problem.

The outer wall of the reactor is fixed, whereas the inner one moves with an azimuthal velocity component  $r\omega(t)$ . The angular speed waveform at the inner wall is modulated, Fig. 3.

This signal can be pulsating, with bias  $\omega_0$ , amplitude  $\Delta\omega$ , positive semi-period  $T_+$  and period  $T$ ; the duty cycle of the waveform is  $D = T_+/T$ , and it represents the time that the waveform assumes a maximum value, bounded as  $0 < D \leq 1$ . This duty cycle is used to constrain both the amplitude and the offset:

$$\Delta\omega = \omega_0 = \frac{\omega_b}{2D} \quad (8)$$

where  $\omega_b = 500$  rpm is the average value of angular speed. This choice ensures that the waveform is either positive or null. Moreover, it maintains the same average value for waveforms with variable duty cycles.

Sigmoid functions were used to smooth the transitions of the angular speed waveform  $\omega(t)$  with a transition duration of 0.05 s to avoid abrupt changes and infinite angular acceleration values, which are impossible to implement in a practical setup.

Fully developed laminar flow Eqs. (9) and (10) were used as inlet conditions. Finally,  $p = 0$  was used as an outlet condition for fluid transport.

$$\mathbf{u} \cdot \mathbf{t} = 0 \quad (9)$$

$$Q_0 = - \int_{\partial\Omega_{int}} 2\pi r \mathbf{u} \cdot \mathbf{n} dS \quad (10)$$

$$[-p\mathbf{I} + \mathbf{K}]\mathbf{n} = -\hat{p}_0\mathbf{n} \quad (11)$$

$$\hat{p}_0 \leq 0 \quad (12)$$

where  $\mathbf{n}$  is the surface normal vector and  $\mathbf{t} = (n_z, -n_r)$  for axial symmetry.

The above model was coupled with the transport of diluted species Eq. (13) and Fick's Law, Eq. (14):

$$\frac{\partial c_i}{\partial t} + \nabla \cdot \mathbf{J}_i + \mathbf{u} \cdot \nabla c_i = R_i \quad (13)$$

$$\mathbf{J}_i = -\mathcal{D}_i \nabla c_i \quad (14)$$

where  $R_i$  is the total rate of reaction of species  $i$  in the solution. The outer wall of the cylinder was considered as a catalyst. At this boundary,

a fast irreversible surface reaction occurs, i.e., the reaction rate is limited by the system's capacity to transport the reactant to the catalytic surface. The reactive system assumes that the reaction rate is infinitely fast, thus yielding a null concentration of the limiting reactant at the reactive wall:  $c_{lim} = 0$ .

By converse, the inner wall of the cylinder is set to a null flux condition  $-\mathbf{n} \cdot \mathbf{J}_i = 0$ . All simulations consider the already defined inlet concentration  $c_0$  as an inlet boundary condition, whereas the outlet (see Fig. 2), was considered as a boundary with outflow condition  $\mathbf{n} \cdot \mathcal{D}_i \nabla c_i = 0$ .

### 2.3. Model meshing

The system described above corresponds to a Graetz-L  v  que-type problem of a specified concentration of a chemical species at the outer wall boundary. Specifically, this problem solves mass transfer due to a fast and irreversible reaction. In these systems, particular attention should be paid to mesh refinement at the reactive wall. A mesh independence test was carried out to assess this by considering multiple wall refinements to evaluate the accuracy of the results obtained from the numerical model. A streamlined diffusion stabilization method was used to improve the overall convergence while avoiding fluctuations of the concentration profile derived from an artificial diffusion method. Mesh independence was tested by evaluating the average concentration at vertical coordinates 0 and  $-H/4$ . Refinements consisted of a combination of quadrilateral elements near the reactor walls and triangular elements elsewhere. Tests consider constant and pulsatile operations of the reactor with a bias angular speed of 500 rpm. The results of this test are shown in Fig. 4.

When using 17,566 elements, the concentration becomes independent of the mesh size. The maximum difference between the average concentration from the last refinement (23,541 elements) and the selected mesh (17,566 elements) is 8 % for the worst case (pulsatile operation at  $-H/4$ ). The other tests yield a difference below 5.6 %. Moreover, the chosen refinement achieves acceptable concentration values at the probe points while being very affordable from a computational perspective. Each simulation under this setting takes approximately two hours.

### 3. Results and discussion

The TCR model was first simulated at different constant angular speeds. Subsequently, a significant portion of the parameter modulation space was explored by simulating the described model for different angular speed waveforms. Specific signals were generated with a matrix

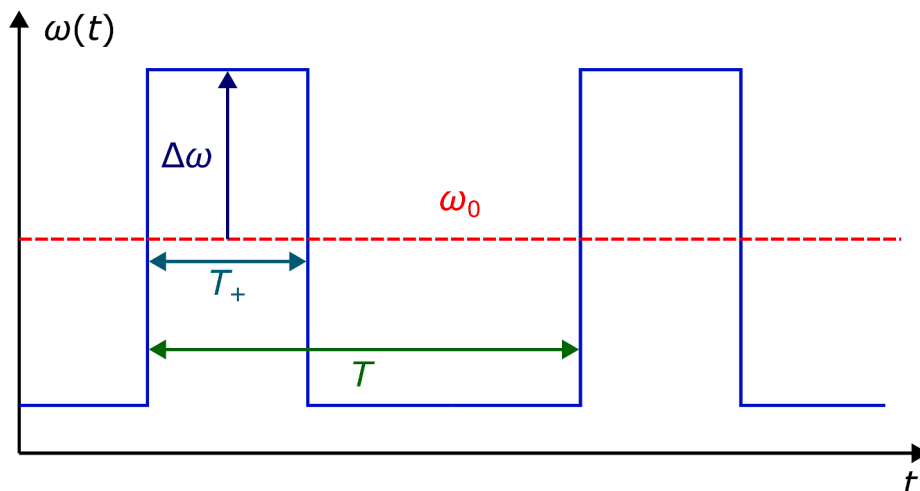


Fig. 3. Inner shaft angular speed profile  $\omega(t)$  defined by bias  $\omega_0$ , amplitude  $\Delta\omega$ , period  $T$ , and positive semi-period  $T_+$ .

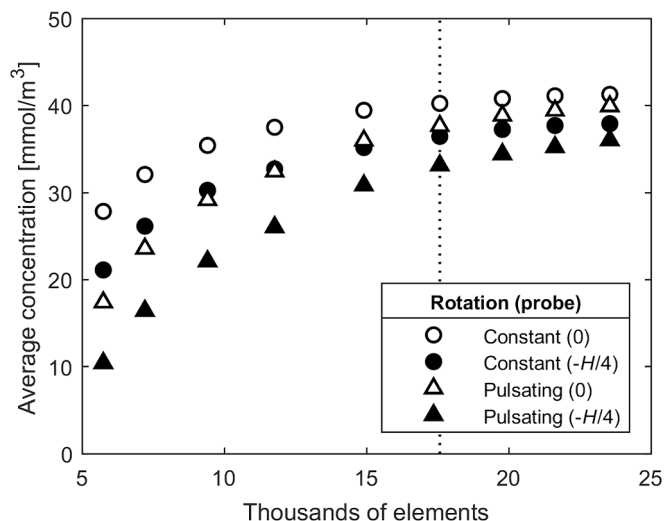


Fig. 4. Mesh sensitivity analysis for two different probe points at (0) and  $(-H/4)$ .

of six period values ( $T$ ) ranging from 5 to 30 s and five duty cycle values ( $D$ ) between 20 and 100 % for 30 runs. These models were executed on a workstation equipped with an AMD Ryzen 9 3950X with 16 cores, a base clock frequency of 3.5 GHz, and 64 GB of RAM. Batch computing was exploited to use multiple cores simultaneously. Additional simulations were run to explore more extended modulation periods at a duty cycle  $D = 20$  %.

### 3.1. Constant angular speed approach

The performance of a catalytic-wall TCR under conventional operating conditions (constant angular speed) was defined as a reference scenario. Fig. 5 shows the behavior of the instantaneous reaction in terms of the overall conversion of the reactant for different angular speeds, corresponding to a  $Re_{rot} \in [471, 3927]$  and equivalent to  $Ta \in [4.9 \cdot 10^4, 3.4 \cdot 10^6]$ . Higher conversion values are observed with an increase in angular speed. This dependence was also observed for catalytic systems, such as the photocatalytic degradation of organic pollutants [26,27]. Fig. 5 also shows that conversion improves linearly with angular speed from 300 to 800 rpm, achieving 26 % to 64 %

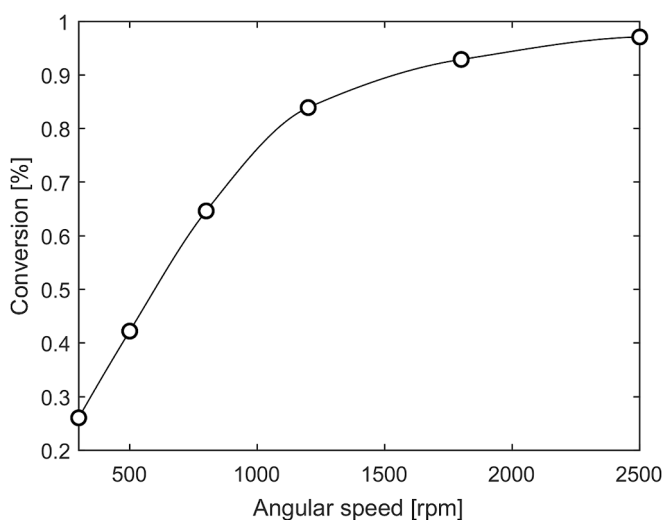


Fig. 5. Average output conversion of the reactor at different constant angular speeds of the inner shaft. For visualization purposes, simulation points (circles) are connected by a shape-preserving interpolant (solid line).

conversions. However, an inflection point in conversion happens at approximately 1000 rpm. This could be attributed to varying flow patterns occurring at different angular speeds.

A change in the flow pattern can be defined in terms of the ratio of rotational Reynolds numbers,  $Re_{rot}^* = Re_{rot}/Re_{rot,cr}$ , where  $Re_{rot,cr}$  is defined as the critical rotational Reynolds number, represented by the analytical expression developed by Esser and Grossmann [28]:

$$Re_{rot,cr} = \frac{1}{0.1556^2} \frac{(1+\lambda)^2}{2\lambda\sqrt{(1-\lambda)(3+\lambda)}} \quad (15)$$

The obtained  $Re_{rot,cr} \sim 96$  agrees with the calculations and observations by Dusting and Balabani [29].

Table 2 shows the estimated  $Re_{rot}^*$  for the different angular speeds tested.  $Re_{rot}^* < 1$  indicates a Laminar Couette Flow (LCF) pattern in the system. The first flow instability happens when  $Re_{rot}^* \sim 1$ , where the flow pattern changes from LCF to Taylor Vortex Flow (TVF). This type of flow indicates an increase in mixing due to the formation of Taylor vortices and thus, a slight increase in conversion. As the angular speed increases, an instability of the TVF occurs yielding a flow pattern characterized by the oscillation of vortices across the azimuthal plane (Wavy Vortex Flow, WVF), further increasing the system's capacity to transfer mass to the catalytic wall. This flow pattern could appear at  $Re_{rot}^* \sim 6$ .

These results agree with the numerical and visualization experiments in [30–32]. Therefore, an increase in conversion from 26 % to  $\sim 75$  % coincides with the change from TVF ( $Re_{rot}^* = 4.91$ ) to WVF ( $Re_{rot}^* = 14$ ).

Moreover, at  $15 < Re_{rot}^* < 22$  a third flow instability could occur (Modulated Wavy Vortex Flow, MWVF). This flow pattern could be seen as a transition pattern, in which the amplitude of the waves formed is modulated by additional frequencies until the first turbulent cells appear at  $Re_{rot}^* \sim 33$ . Finally, fully Turbulent Taylor Vortex Flow (TTVF) could appear at  $Re_{rot}^* > 35$ .

### 3.2. Pulsating angular speed approach

Fig. 6 shows the results from the parametric study of different modulation profiles of angular speed. A duty cycle of 100 % for all the periods represents the operation at constant, nominal angular speed (500 rpm). Combinations of duty cycles and periods denote different angular speed profiles, as described by Eq. (8). Conversion increases with a reduction in duty cycle, leading to a maximum 57.2 % at  $D = 20$  % and  $T = 30$  s. This behavior becomes relevant, as a modulated rotational pattern can be exploited to actuate the reactor for brief time instants while keeping it idle during longer, complementary phases (see Fig. 7).

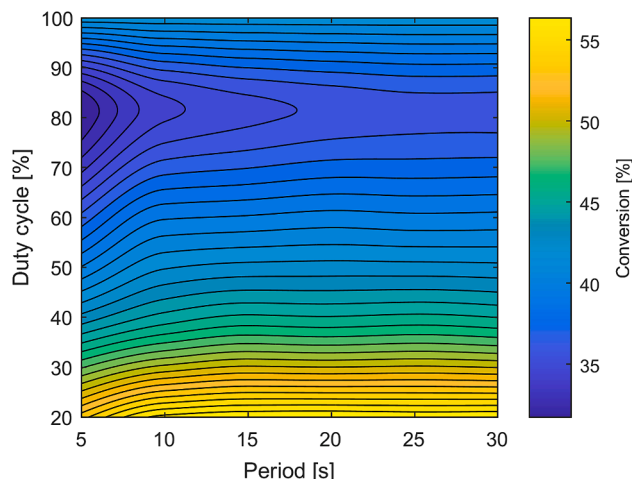
These results agree with similar observations of electrochemical reactors, where modulated input signals are applied [33,34]. It is noteworthy that the modulated profile strategy shows a local minimum at  $D \sim 80$  %. In this condition, the flow patterns for both constant and modulated operations fall within the WVF regime. Thus, the pulsating modulation does not bring any advantage to enhance conversion and compensate for idling operation. In contrast, modulations where  $D < 60$  % can induce a further dynamic change in the flow pattern (MWVF

Table 2

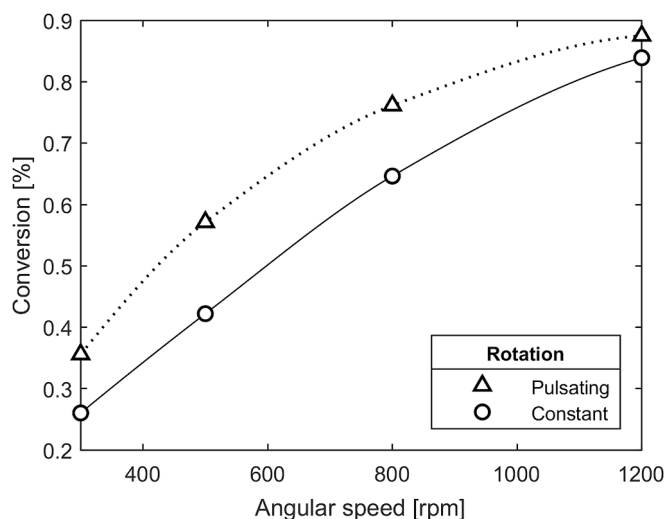
Estimated rotational Reynolds numbers for simulated constant angular speed cases.

| $\omega$ [rpm] | $Re_{rot} [-]$ | $Re_{rot}^* [-]$ |
|----------------|----------------|------------------|
| 300            | 471            | 4.91             |
| 500            | 785            | 8.19             |
| 620            | 974            | 10.15            |
| 712            | 1118           | 11.66            |
| 830            | 1304           | 13.59            |
| 1000           | 1571           | 16.37            |
| 1250           | 1963           | 20.47            |
| 1667           | 2619           | 27.29            |
| 2500           | 3927           | 40.94            |





**Fig. 6.** Average output conversion map with varying angular speeds for the inner shaft. Angular speed waveforms are defined by their duty cycle and period.



**Fig. 7.** Average output conversion with different nominal angular speed values, applying pulsating and constant modulations. Pulsating profiles were obtained with a duty cycle of 20 % and a period of 30 s.

and transition to TTVF at  $D < 20$  %).

For periods above 15 s, conversion appears to become independent of the modulation period. Table 3 gives a closer insight into this trend. An increase in the modulation period yields a slight reduction in the average output concentration and, thereby, a subtle improvement in the average conversion. However, as the period escalates, the output concentration response's root-mean-square (RMS) value increases. Note that, for a constant signal of output concentration, average and RMS values in the time domain should coincide. Hence, favorable conversion comes at the cost of considerable variability. This is physically explained

**Table 3**

Average and root-mean-square concentration values with a duty cycle of 20 % and different periods.

| T[s] | avg(C) [mmol/m <sup>3</sup> ] | rms(C) [mmol/m <sup>3</sup> ] | rms(C) – avg(C) [μmol/m <sup>3</sup> ] |
|------|-------------------------------|-------------------------------|----------------------------------------|
| 15   | 21.489                        | 21.497                        | 8.74                                   |
| 20   | 21.476                        | 21.486                        | 9.91                                   |
| 25   | 21.436                        | 21.448                        | 12.22                                  |
| 30   | 21.391                        | 21.406                        | 14.93                                  |

by the fact that the reactor's dynamics can follow slower modulation signals more effectively. Then, the variable nature of the modulation waveform is more evident in the output concentration when the period increases. This property is exploited similarly in electrical circuits with the pulse width modulation technique.

In terms of amplitude, signals where the duty cycle tends to zero appear to improve conversion. Nevertheless, this decrease finds a limitation in the system's mechanics, as small values of  $D$  could lead to excessive angular speeds, according to Eq. (8).

Fig. 7 compares pulsating and constant operations of the TCR at different nominal angular speeds. As discussed above, for the constant operation, the increase in conversion is related to a change in flow patterns. Each angular speed during constant operation is specifically related to a flow. In contrast, the higher conversion values obtained for the pulsatile operation are related to a dynamic change in the flow pattern for each nominal angular speed value  $\omega_b$ . That is, fluid is changing from an axially dispersed laminar flow (ADF) profile to MWVF for each  $\omega_b$  point. Increasing this value during pulsatile operation is related to higher  $Re_{rot}^*$  numbers and thus, longer operation periods that could reach a fully developed TTVF.

These phenomena are illustrated in Fig. 8 and Fig. 9. They show the dynamic effect of the inner shaft actuator on the velocity (streamlines) and concentration profiles at the first half of the TCR ( $+H/2$ ) with a duty cycle of 20 % and period of 30 s. During idle operation, the TCR performance is limited by the effective amount of reactant that reaches the catalytic wall; it is mainly controlled by radial diffusion and dispersion (ADF) at  $Re_{ax}^* = 33$ , Fig. 8 ( $t = 157.4$  s).

Flow instabilities developed with a sudden increase in the angular speed (from idle to active), leading to a change in the flow pattern of the reactor. A velocity profile disruption occurs from 157.6 to 157.9 (Fig. 8), with an associated decrease in the local concentration. This is a consequence of the Taylor vortices' dynamic formation and axial propagation. It is worth highlighting this propagation effect on the modulated fluid velocity profile in the reactor. Two phenomena can be observed at 157.6 s: first, flow instability and formation of TVF near the inlet of the reactor ( $Re_{rot}^* > 1$ ); second, LCF ( $Re_{rot}^* < 1$ ) at the middle section of the reactor. This propagation and flow perturbation happens until it reaches a quasi-steady state at 158 s, in which a stable MWVF pattern is observed throughout the reactor ( $Re_{rot}^* > 22$ ).

A similar but reversed pattern is shown in Fig. 9, where the reactor transitions from active to idle condition. A drop of 40 % in the local concentration occurs during this transition in the flow pattern (at  $t = 163.5$  s,  $+H/2$ ). Finally, from 163.6 s to 165 s, a transition from WVF to LCF and then to ADF occurs due to an abrupt stop of the angular speed. A local increase in concentration is expected since the capacity for mass transfer decreases.

### 3.3. Mixing profiles

A mixing index,  $I_{mix}$ , is defined to understand both constant and pulsating angular speed approaches through the mass-transfer capabilities of the reactor, Eq. (16).  $I_{mix}$  is defined by the normalized mixing cup concentration,  $C_{cup}$ , Eq. (17), and the variance of the normalized concentration profile between a pair of vortices tracked along the TCR.

$$I_{mix} = \frac{\sigma^2}{C_{cup}(1 - C_{cup})} \quad (16)$$

$$C_{cup} = \frac{\int c v dr dz}{c_0 \int v dr dz} \quad (17)$$

During the formation of TVF to MWVF patterns,  $C_{cup}$  was computed regarding the azimuthal velocity profile ( $v$ ) of the chosen tracked vortices as they flow through the reactor. This value was determined as a function of the radial ( $r$ ) and axial ( $z$ ) velocity profiles by fixing the area of the tracked vortices and displacing it at a constant axial velocity.

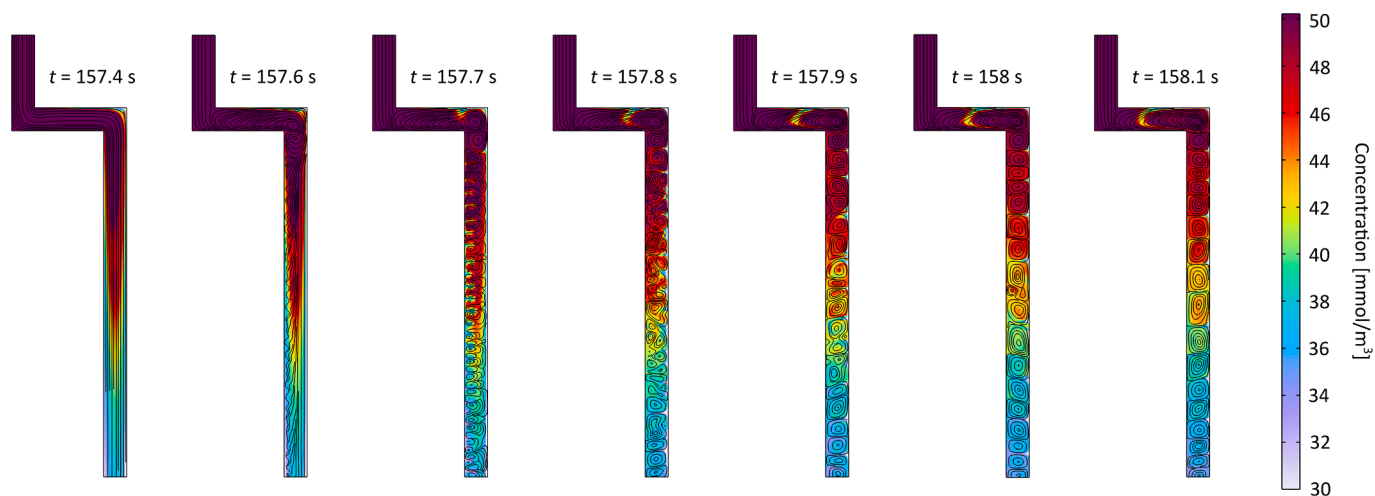


Fig. 8. Velocity streamlines and concentration profiles with a duty cycle of 20 % and period of 30 s in the region  $+H/2$ . Transition from idle to active condition.

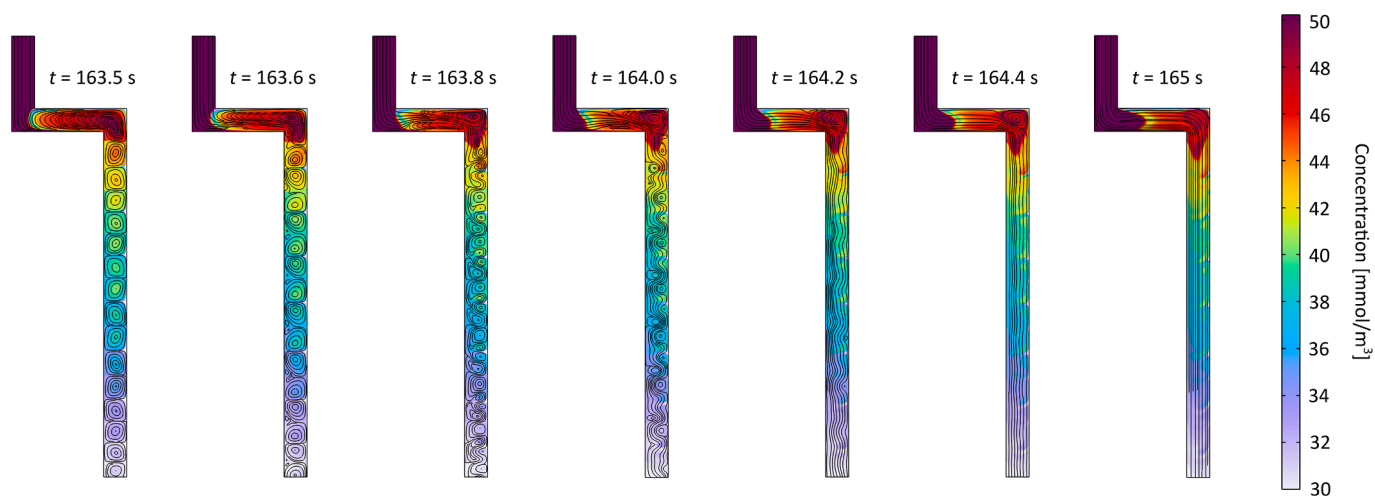


Fig. 9. Velocity streamlines and concentration profiles with a duty cycle of 20 % and period of 30 s at the region  $+H/2$ . Transition from active to idle condition.

COMSOL Multiphysics simulation data were post-processed using the Livelink tool for MATLAB. At each point, vortices were identified by finding concentration peaks along  $z$  in the middle ring. Two vortices were isolated and tracked down the reactor. Discrete calculations were carried out at each time instant.

Fig. 10(a) shows the mixing index for both constant and pulsating angular speed actuation, while Fig. 10(b) shows the velocity streamlines and concentration profiles associated with the actuation.  $I_{mix}$  results for constant angular speed follow previously reported results [29].

Fig. 10 illustrates three colored regions corresponding to the dynamics of the TCR. Region 1 relates to the sudden increase in angular speed, Region 2 indicates the abrupt stop of the inner shaft, and Region 3 matches the TCR operation under ADF.

In Region 1, the sudden increase in angular speed leads to a sharp decrease (70.3 %) of  $I_{mix}$ , implying higher mixing capacity during the chaotic formation of the vortices and their propagation, and higher mass transfer rates to the catalytic wall. The sudden drop followed by an increase of  $I_{mix}$  within Region 1 indicates the dynamics of vortex formation. When the MWVF is reached, a constant decrease of  $I_{mix}$  occurs similarly to the behavior under constant angular speed. Region 1 highlights the enhancement in the mass transfer capacity of the actuating signal. Additionally, the magnitude of  $I_{mix}$  decreases as the reaction progresses (limiting the mass transfer rates in the system). Mass transfer at a normalized time of 0.8 reaches an asymptotic value; this limitation

is also observed for the constant angular speed signal.

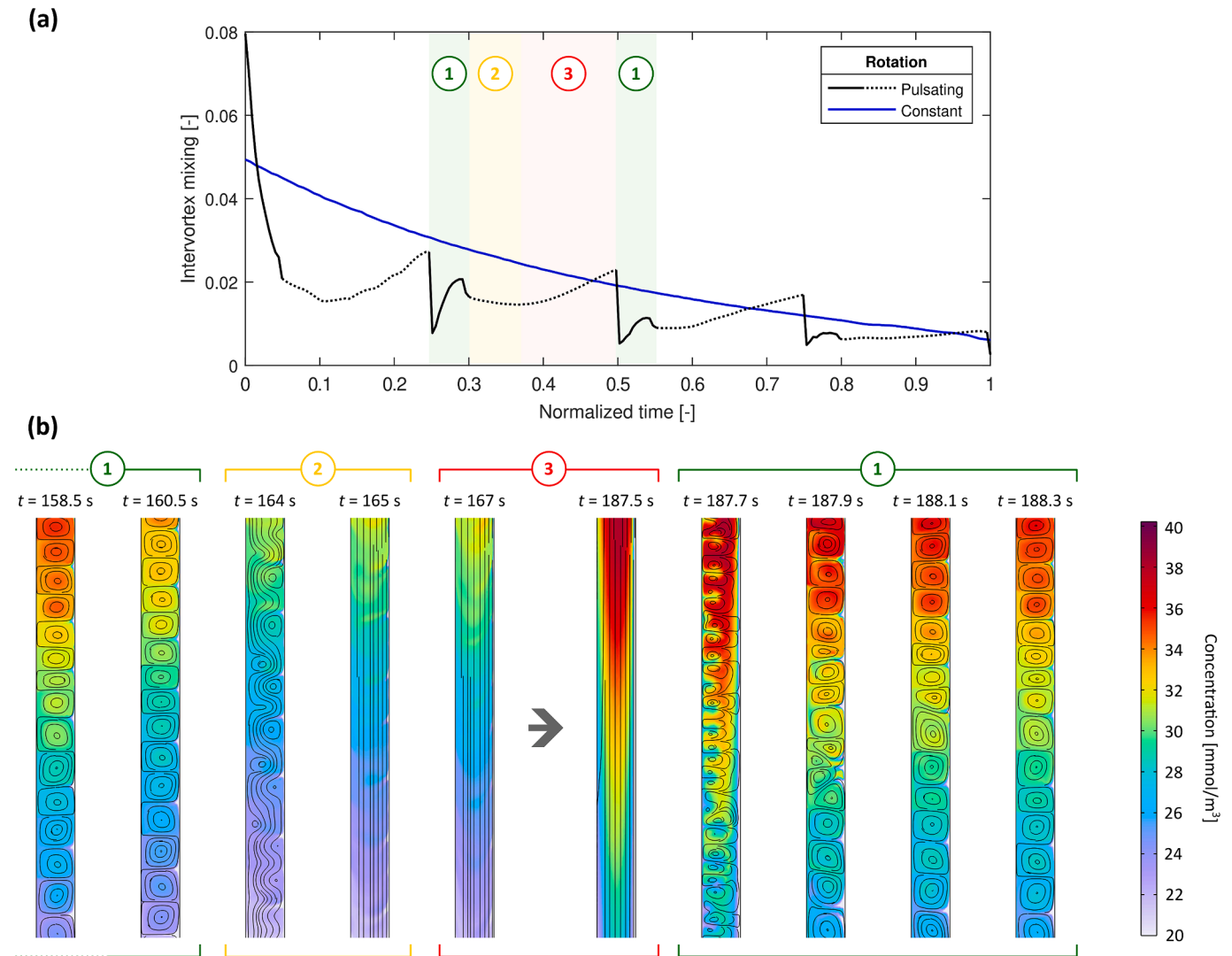
Region 2 can be considered a transitory regime in which the inner shaft is suddenly stopped, and the flow pattern changes from MWVF to ADF. The chaotic destruction of the formed vortices and their inertia provide a certain degree of mixing since  $I_{mix}$  slowly decreases (9.5 %). The stagnant regions of the vortices, where higher concentrations are locally observed, are altered, thus favoring a homogenization of the concentration profile.

As soon as a fully developed laminar flow is established, axial dispersion and radial diffusion control the system's capacity for mass transfer; this corresponds to Region 3. The reduction in mass transfer from the bulk to the reacting wall, combined with the dispersion of unreacted components, leads to an increase in  $I_{mix}$  (54.3 %). However, this increase is abruptly stopped by the growth in the angular speed of the reactor. The complete evolution of the concentration and velocity profiles within these regions is observed in Fig. 10(b).

These are clear benefits of the effect of a modulated signal on the operation of the TCR in terms of mixing and mass transfer on the overall conversion achieved: 57.2 % compared to 42.2 % under constant modulation.

#### 3.4. Power consumption

The electric motor that drives the reactor introduces an input torque



**Fig. 10.** (a) Evolution of the mixing index for constant and pulsating (duty cycle 20%) angular speeds at the inner shaft of the Taylor-Couette reactor. (b) Velocity streamlines and concentration profiles are also illustrated.

waveform given by

$$T_{mot} = \underbrace{J_{tot} \frac{d\omega}{dt}}_{T_{in}} + \underbrace{T_0 \text{sign}(\omega)}_{T_{bear}} + \underbrace{\beta_2 \omega^2 \text{sign}(\omega) + \beta_1 \omega}_{T_{drag}} \quad (18)$$

The first term accounts for the inertial contribution of the inner shaft of the reactor and the electric machine, as  $J_{tot} = 1.66 \cdot 10^{-4} \text{ kg/m}^2$  is the moment of inertia of the rotating parts. The second contribution is the friction torque of the bearings, where  $T_0 = 0.1 \text{ Nm}$ . The drag torque due to viscous and pressure forces is computed from the CFD model in COMSOL Multiphysics through

$$T_{drag} = 2\pi r_i^2 \int_H p dl \quad (19)$$

and is approximated by a second-degree polynomial, where  $\beta_2 = 7.19 \cdot 10^{-8} \text{ Nms}^2/\text{rad}^2$  and  $\beta_1 = 1.51 \cdot 10^{-5} \text{ Nms/rad}$ .

A Maxon EC-i 52 brushless permanent-magnet motor is proposed to account for the electric machine. It presents a back electromotive force and torque constant  $K_e \cong K_t = 93.1 \text{ mNm/A}$ . Following a direct-current motor equivalent representation, the motor current can be computed as

$$i_{mot} = \frac{T_{mot}}{K_t} \quad (20)$$

and thus, the voltage applied to the motor can be determined through the armature circuit equation:

$$e_{mot} = L \frac{di_{mot}}{dt} + Ri_{mot} + K_e \omega \quad (21)$$

where  $L = 0.28 \text{ mH}$  and  $R = 0.178 \Omega$  are the motor winding inductance and resistance, respectively. The motor's electrical power can be computed as

$$P_{mot} = e_{mot} i_{mot} \quad (22)$$

This power is further corrected with the efficiency of the motor controller  $\eta_{cont} = 90 \%$ :

$$P_e = \begin{cases} P_{mot}/\eta_{cont} & \forall P_{mot} \geq 0 \\ P_{mot}\eta_{cont} & \forall P_{mot} < 0 \end{cases} \quad (23)$$

Average electrical power consumption was computed for both modulation profiles: constant and pulsating. In this way, the reactor power economy can be computed for the operating points depicted in Fig. 7. The objective is to compare the constant-rotation case against the modulated one with a duty cycle of 20 % and a period of 30 s. Both systems must be evaluated at the same output conversion points to obtain a fair comparison. For instance, a constant angular speed of 689.2 rpm yields the same average output conversion as a pulsating



angular speed waveform with  $\omega_b = 500$  rpm, duty cycle of 20 %, and period of 30 s.

Fig. 11 highlights the average consumed power behavior of the TCR for both constant and pulsating rotation driving strategies. This comparison is done by interpolating the conversion characteristics in Fig. 7 with a piece-wise cubic function.

It is seen that, for the studied TCR configuration, pulsating modulation yields lower average power consumption for conversions ranging from 35.6 % to 80.8 %. The best operating point in terms of power saving is found at a conversion of 54.9 %, where the pulsating waveform yields a power consumption improvement of 25 % compared to the constant modulation counterpart. After a conversion of 80.8 %, pulsating speed profiles are no longer beneficial in terms of power, as the attained speeds induce larger power losses in the system. This effect is mainly driven by the quadratic drag torque behavior.

#### 4. Conclusions

This work analyzes the use of Taylor-Couette reactors (TCR) as an intensified platform to test the effect of the periodic modulation of angular speed on the reactor performance. A simulation of a catalytic wall TCR was performed in different conditions of modulation of the reactor inner shaft: (i) constant angular speed with  $Re_{rot} \in [471, 3927]$ , and (ii) angular speed waveforms characterized by different duty cycles and periods. The operation at a constant angular speed of 500 rpm yields a conversion of  $\sim 42.7$  %. In contrast, when applying a pulsating modulation of the same average speed, a conversion of 57.2 % is attained with a duty cycle of 20 % and a period 30 s. This type of operation achieves a higher conversion than the constant angular speed approach with 80 % of idle time. Additionally, this type of TCR operation reduces average power consumption by 25 %. The conversion improvement and reduction in the power consumption were achieved under highly limiting conditions for mass transfer ( $Pe \sim 3 \cdot 10^5$  and  $Sc \sim 1 \cdot 10^5$ ). The modulated strategy could potentially yield better results also in more favorable scenarios.

The obtained results can be explained in terms of the change in the mixing regimes or regions due to the dynamic evolution of the flow patterns in the TCR. The modulation of the angular speed directly impacts the mixing index and, thus, the mass-transfer capability of the TCR system. An abrupt drop in the mixing index of 70.3 % is related to a sudden increase in the angular speed, inducing different flow perturbations by changing from an axially dispersed laminar flow to a wavy vortex flow. The magnitude of these abrupt drops in the mixing index can be analogously viewed as a local increase in the mass-transfer coefficient until an asymptotic value is reached.

The use of modulated input signals opens the possibility of improving intensifying equipment. At the same time, this research project focuses on an on-off input signal; future efforts could address the use of different signals, optimization techniques, and feedback control strategies to increase the benefits of this type of operation. The integration of signal modulation is a fundamental part of the philosophy behind Process Intensification 4.0.

#### CRediT authorship contribution statement

**Enrique A. López-Guajardo:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Renato Galluzzi:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Fernando Delgado-Licona:** Writing – original draft, Visualization, Investigation, Formal analysis. **Ruben Morales-Menendez:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Funding acquisition.

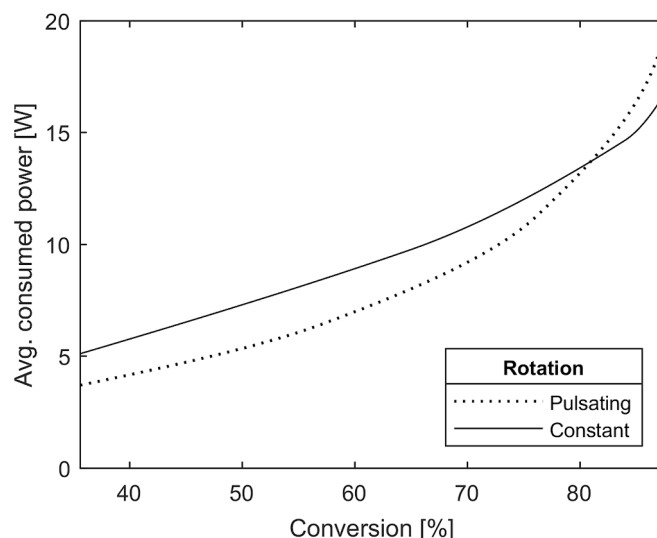


Fig. 11. Average consumed power for pulsating and constant modulations. To yield a fair comparison, power consumption is compared at nominal speeds where both modulation profiles yield the same average output conversion.

#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

#### Data availability

Data will be made available on request.

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